



Project Specification Document: CanvasAI

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Track: AI for Inclusive & Accessible Opportunities

Tagline: Engineering the shift from passive video broadcasting to a stateful, interactive visual tutoring engine.

1. Executive Summary

Online professional education is bottlenecked by passive, one-size-fits-all video lectures. While effective for high-level theory, video fails at teaching complex, "inside-out" engineering mechanics (e.g., pointer manipulation, memory management, deep DSA). If a learner asks an edge-case question, static video cannot adapt. CanvasAI replaces the video player with a dynamic, stateful orchestration engine. It translates pedagogical logic and uploaded technical documents into structured spatial data, rendering a live, interactive React Flow whiteboard where concepts are animated in real-time.

2. Core Feature Specifications

- **Interactive React Flow Sandbox:** A fully draggable, zoomable visual canvas. Users can physically interact with the data structures (e.g., clicking a node to delete it) to trigger real-time LLM recalculations and visual error simulations (like memory leaks or segmentation faults).
 - **Stateful Context Reversion ("The Time Machine"):** A granular "Slide Back" system. If a user follows a conversational tangent, they can revert to a previous state. The frontend instantly redraws the older graph, and the backend purges the tangent from the LLM's context window to prevent context drift.
 - **Document-to-Diagram (Visual RAG):** Users can upload complex academic papers or API documentation. The system embeds the text and performs high-dimensional similarity searches to ground the AI's visual explanations strictly in the provided text, entirely mitigating hallucination.
 - **Cognitive Load Adaptation:** Natively supports neurodivergent learning profiles by adjusting pacing and rendering. A "High-Stimulation" profile utilizes rapid micro-steps, while a "Dyslexia" profile emphasizes spatial mapping, custom node shapes, and color-coding over dense text blocks.
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3. Technical Architecture (The Alpha Prototype Stack)

CanvasAI utilizes a decoupled, asynchronous architecture designed for zero-latency state updates and robust multi-agent orchestration.

A. The Frontend Layer (React + React Flow)

- **Framework:** React.js optimized for single-page application (SPA) performance.
- **Visualization Engine:** React Flow. Instead of handling static SVG generation, the frontend reads strict JSON payloads (`nodes` and `edges`) from the WebSocket to mount, unmount, and animate complex graphs dynamically.
- **Custom Components:** Implementation of specialized React Flow nodes (e.g., "Memory Block Node," "Stack Frame Node") to visually represent low-level hardware/software interactions.

B. The Orchestration Layer (FastAPI + WebSockets)

- **Protocol:** Persistent WebSockets via FastAPI to stream text explanations character-by-character while simultaneously pushing JSON state-updates to the React canvas without HTTP polling latency.
- **The Multi-Agent Pipeline (LangChain):**
 1. **Agent 0 (The Retrieval Engine):** Triggers when a document is uploaded. It handles chunking and queries the vector database to inject factual context into the system prompt.
 2. **Agent 1 (The Context Synthesizer):** Analyzes the user's vague prompt against the current visual state of the canvas, translating intent into a precise technical directive.
 3. **Agent 2 (The Pedagogical Architect):** Determines the instructional strategy and formulates a visual script (e.g., *"Highlight the target node in red, then reroute the pointer edge"*).
 4. **Agent 3 (The Schema Enforcer):** A rigid formatting model. It takes the visual script and strictly outputs the minified `{"nodes": [...], "edges": [...]}` JSON payload required by React Flow, actively rejecting conversational tokens to prevent UI crashes.

C. The Semantic Data Layer (Supabase)

- **State Logging (`JSONB`):** Every conversation turn and corresponding React Flow JSON state is logged sequentially in a PostgreSQL `JSONB` column. This serves as the immutable ledger for the "Time Machine" reversion feature.
- **High-Performance Vectorization (`pgvector`):** Document embeddings are stored natively. For optimal retrieval speed during Visual RAG operations, the architecture is designed to support advanced similarity search optimization, evaluating indexing techniques (such as benchmarking FAISS workflows against standard binary vectorization) to handle dense algorithmic texts without bottlenecking the real-time drawing process.

4. Workflow Examples

Workflow 1: The "Break the State" Interaction

1. **State:** The canvas displays a functioning Binary Search Tree.

2. **Trigger:** User asks, *"What happens if I insert a duplicate value?"*
3. **Execution:** FastAPI receives the socket event. Agent 1 identifies the context. Agent 2 dictates the error state. Agent 3 generates the updated JSON.
4. **Resolution:** The React Flow canvas dynamically animates a new node attempting to attach, flashing red, and displaying an error badge based on the new JSON payload.

Workflow 2: The "Time Machine" Reversion

1. **State:** User is 10 steps deep into a tangent about hash collisions.
 2. **Trigger:** User clicks "Revert to Step 4".
 3. **Execution:** React instantly fetches the Step 4 `JSONB` payload from Supabase and redraws the canvas.
 4. **Backend Adjustment:** FastAPI intercepts the next user message, queries Supabase for the conversation history up to Step 4 only, and dynamically rebuilds the LLM's context window. Steps 5-10 are permanently pruned from the active memory.
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5. Business Impact for LinkedIn

CanvasAI transitions LinkedIn Learning from a passive content repository into an active, verifiable training sandbox. By automating high-fidelity visual tutoring and integrating custom document processing, it vastly improves completion rates for highly technical subjects while setting a new accessibility standard for diverse learning styles in enterprise education.